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CASE REPORT

Pseudoaneurysm of the Superficial Temporal Artery: A Complication of Botulinum Toxin Injection

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Abstract Pseudoaneurysms of the superficial temporal artery (STA) must be considered in the differential diagnosis of masses of the lateral forehead and temporal fossa. Although the first reported case of a temporal artery aneurysm was by Thomas Bartholin in 1740, this lesion receives scant mention in the plastic and maxillofacial surgical literature. A history of recent blunt trauma or surgery to the forehead combined with a pulsatile bruit should direct the physician to this diagnosis. A case of pseudoaneurysm arising from the frontal branch of the left STA 2 months after Botox injection is presented. The patient underwent embolization of the STA with Onnix. A similar case was previously reported once in the world literature.

Keywords Botox · Pseudoaneurysm · Superficial temporal artery · Trauma

Approximately 400 cases of traumatic pseudoaneurysms of the superficial temporal artery (STA) are reported in the literature [1, 2]. Although the mechanism for these pseudoaneurysms has varied from sword lacerations to sports injuries and motor vehicle accidents, they share a characteristic location and clinical picture [2–6]. The usual presentation of this lesion is that of a painful pulsatile mass in the frontotemporal region associated with local fremitus and bruit 2 weeks to 4 months after the initial trauma [1].

Treatments have included embolization, excision, percutaneous suturing, and compression [3, 6]. The indications for surgery include pain, cosmetic deformity, and prevention of spontaneous rupture [7].

This report includes a review of the literature and presents a documented case in which a Botox injection was the cause of an STA pseudoaneurysm.

Materials and Methods

A 60-year-old woman presented to the clinic with a history of a left temporal, nontender, nonmobile, nonerythematous, pulsating mass, which appeared 2 months after a Botox injection for temporal brow-lift and crow's feet correction, with no changes in its size or consistency over time but rather an immediate bluish discoloration after the procedure. The patient had no medical history and denied the use of aspirin or any other anticoagulating medications at the time.

Because the doctor who performed the procedure was an outside physician, the patient reported that she received 4 units of Botox above the eyebrows and 6 units within the left lateral periorbital area. Assuming that the usual dilution is 100 units per 2.5 ml of normal saline or that 100 units per 4 ml of normal saline were used, we concluded that the amount injected was respectively 0.1 or 0.15 ml of Botox.

Results

Magnetic resonance imaging (MRI) of the clinically palpable mass in the temporal fossa exhibited a 1.6-cm lesion

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in the superficial aspect of the left temporal muscle invading the deep aspect of the subcutaneous tissues. The lesion had a multilobulated appearance with clearly defined borders (Fig. 1). Doppler ultrasound showed hypervascularity peripherally, with high flow in the veins. A vascular lesion was suspected.

The patient underwent a cerebral angiogram that demonstrated an aneurysm of the distal frontal branch of the left STA with early venous filling, which favored a traumatic pseudoaneurysm with a direct fistula (Fig. 2). The patient was offered the option of open surgery or endovascular therapy and chose the latter due to cosmetic preference.

Onnix embolization of the feeding artery and microcatheterization of the left STA was performed (Figs. 3, 4). Due to tortuosity of the artery, the Onnix embolization could not be delivered to the aneurysm [8]. The patient was discharged home with good recovery. She was informed that she was prone to recurrence. At this writing, she will be followed up and treated with surgical resection in case of relapse.



Fig. 2 Left carotid artery angiogram showing a 1.6-cm pseudoaneurysm of the frontal branch of the superficial temporal artery (STA) (*arrow*)

Discussion

The first case of temporal artery aneurysm was reported in 1740 by Thomas Bartholin. Most cases (~75%) are the result of blunt head injury [7–10], but there also are cases related to bicycle helmet [7], sword laceration, sports injury, motor vehicle accident [6, 9–12], penetrating scalp lesions [13], external ventricular drainage catheters [1], use of pin-type head-holder devices [14], and previous craniotomy [15].

Injections with botulinum toxin generally are well tolerated and have few side effects [16]. Generalized idiosyncratic reactions are uncommon, generally mild, and transient. The patient may experience mild injection pain and local edema, erythema, transient numbness, headache, malaise, or mild nausea.

A rare complication with pseudoaneurysm of the frontal branch of the STA after Botox injection was described earlier in the literature [17]. This case report describes a patient who initially underwent surgery with a diagnosis of

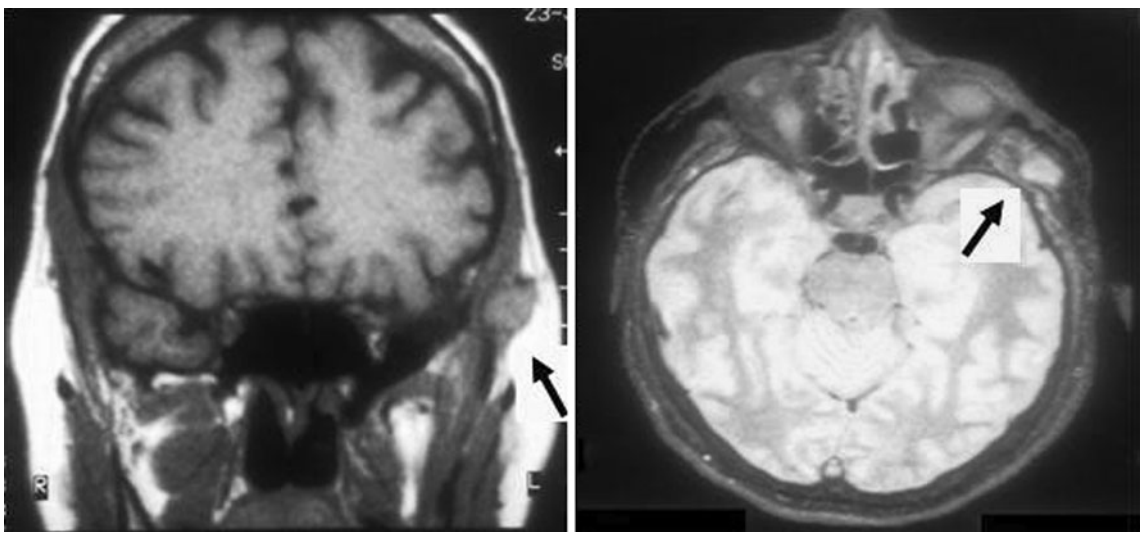


Fig. 1 A 1.6-cm lesion in the superficial aspect of the left temporal muscle (*arrow*). The lesion shows clearly defined borders



Fig. 3 Onnix microcatheter angiogram showing the feeding arteries and the aneurysm (*arrow*)



Fig. 4 Postembolization angiogram showing disappearance of the aneurysm (*arrow*)

a vascular tumor but was found to have a pseudoaneurysm instead, as evidenced by histopathologic analysis of the resected lesion. This pseudoaneurysm was attributed to the history of a 30-ml Botox injection 6 months before the appearance of a right temporal tender mass. We concluded in our case report that the amount of Botox injected was not the offending factor but rather the direct injection into the STA because the patient reported immediate bruising, indicating subcutaneous hematoma.

Pseudoaneurysms can arise in the forehead and scalp due to blunt traumatic impingement of the STA against the calvarium [7, 10]. The anterior branch of the artery is most vulnerable because in the lateral forehead, it courses over the frontal osseous ridge in the galea aponeurotica formed by the fusion line of the deep and superficial temporalis muscle fascia (Fig. 5). This dense fascial investment has a tethering effect in the gap between the temporalis and the frontalis muscles and prevents the artery from displacing laterally in response to traumatic forces.

It is believed that the arterial wall is injured during trauma, or there would be a contusion with subsequent wall necrosis [12]. Blood possibly extravagates from the injured artery, with the formation of a hematoma and a pseudocapsule around it. The hematoma capsule would expand, and the clot would reabsorb, resulting in a cavity leading to pseudoaneurysm formation.

The diagnosis of pseudoaneurysms usually is suspected because of the recent traumatic antecedents and the physical examination. Digital compression of the proximal segment on the STA usually abolishes or decreases the bruit and the fremitus. The differential diagnosis includes simple hematomas, abscess, soft tissue tumors, neuromas

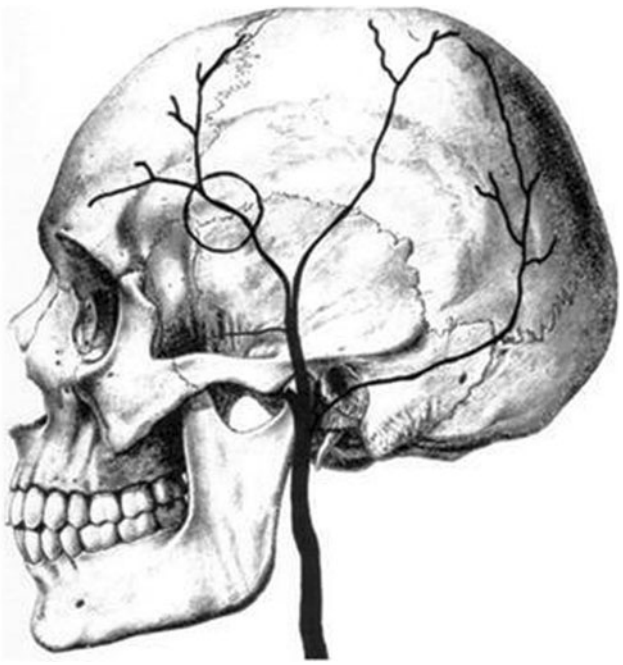


Fig. 5 Locations of the anterior, middle, and posterior branches of the superficial temporal artery as it courses over the temporal ridge at the anterior superior origin of the temporalis muscle. In the *circle area* (superior to the temporalis and lateral to the frontalis muscle), the artery is relatively fixed in the galea aponeurotica (muscle gap) and particularly prone to traumatic lesions and pseudoaneurysm formation [10]

and foreign body granulomas. Complementary studies such as Doppler ultrasound and contrast-enhanced computed tomography (CT) scans may confirm or rule out other lesions mimicking a pseudoaneurysm, but the definitive diagnosis usually is made by angiography [14, 18].

Therapeutic options include surgical resection, ligation, sclerosis, ultrasound-guided thrombin injection, and coil embolization. Studies show that nearly 89% of untreated pseudoaneurysms resolve in 5 to 90 days, but no duplex criteria have proved predictive of spontaneous resolution. Large or expanding pseudoaneurysms should be treated quickly. For symptomatic pseudoaneurysms, duplex-guided compression therapy and thrombin injections are useful. Treatment is indicated to prevent bleeding, relieve symptoms, and favor a desirable cosmetic result. Some authors have reported good results using embolization for the treatment of these lesions, but surgery is the treatment of choice, with ligature of proximal and distal vessels and excision of the pseudoaneurysm [12, 17].

Conclusion

Superficial temporal artery aneurysms resulting from blunt or penetrating trauma have been described previously. Our case report describes an STA aneurysm after Botox injection. A similar case was previously described once in the literature [17]. We attribute this lesion to a technical fault at the time of the injection because no documented anatomic deformity was in evidence but rather bruising after the procedure.

Usually, STA aneurysms occur at the superior temporal line because the anterior arterial branch is uncushioned by a muscle gap and lies directly on the periosteum. The artery also is tethered by the adventitia to the frontalis and temporalis muscles. All these anatomic peculiarities increase the likelihood of a lesion resulting from a blunt trauma [1, 10]. The diagnosis is determined by angiography.

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